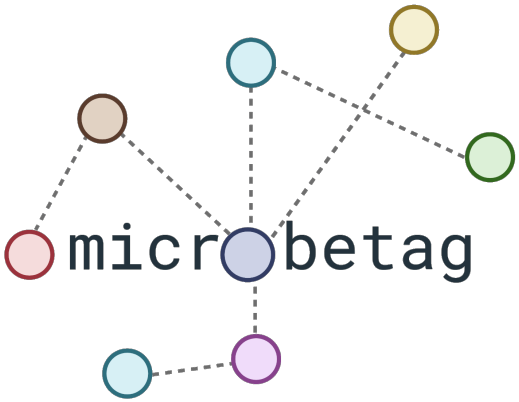
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Gossip in the Microbiome: Metabolic Secrets Behind Microbial Co-Occurrence

HSCBB26

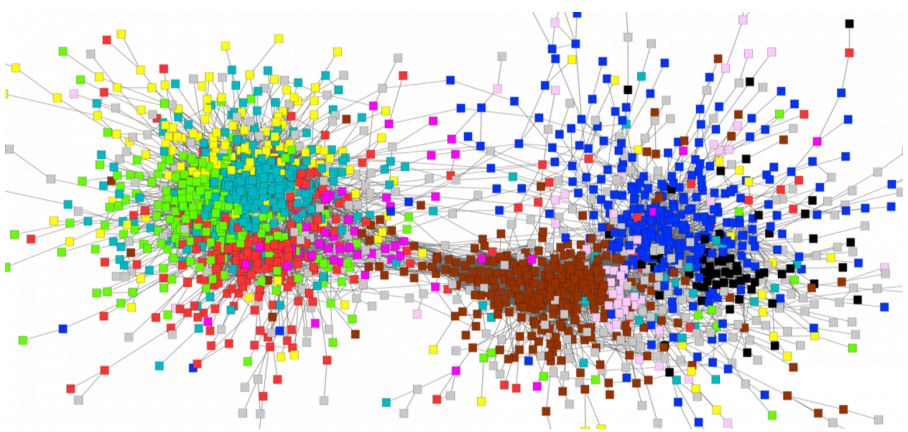
Haris Zafeiropoulos

Post-doctoral researcher
Mechanisms of Microbial Metabolic Interactions (m3i)
Friedrich Schiller University Jena
Cluster of Excellence Balance of the Microverse
Institute of Biodiversity, Ecology and Evolution

hariszaf.github.io

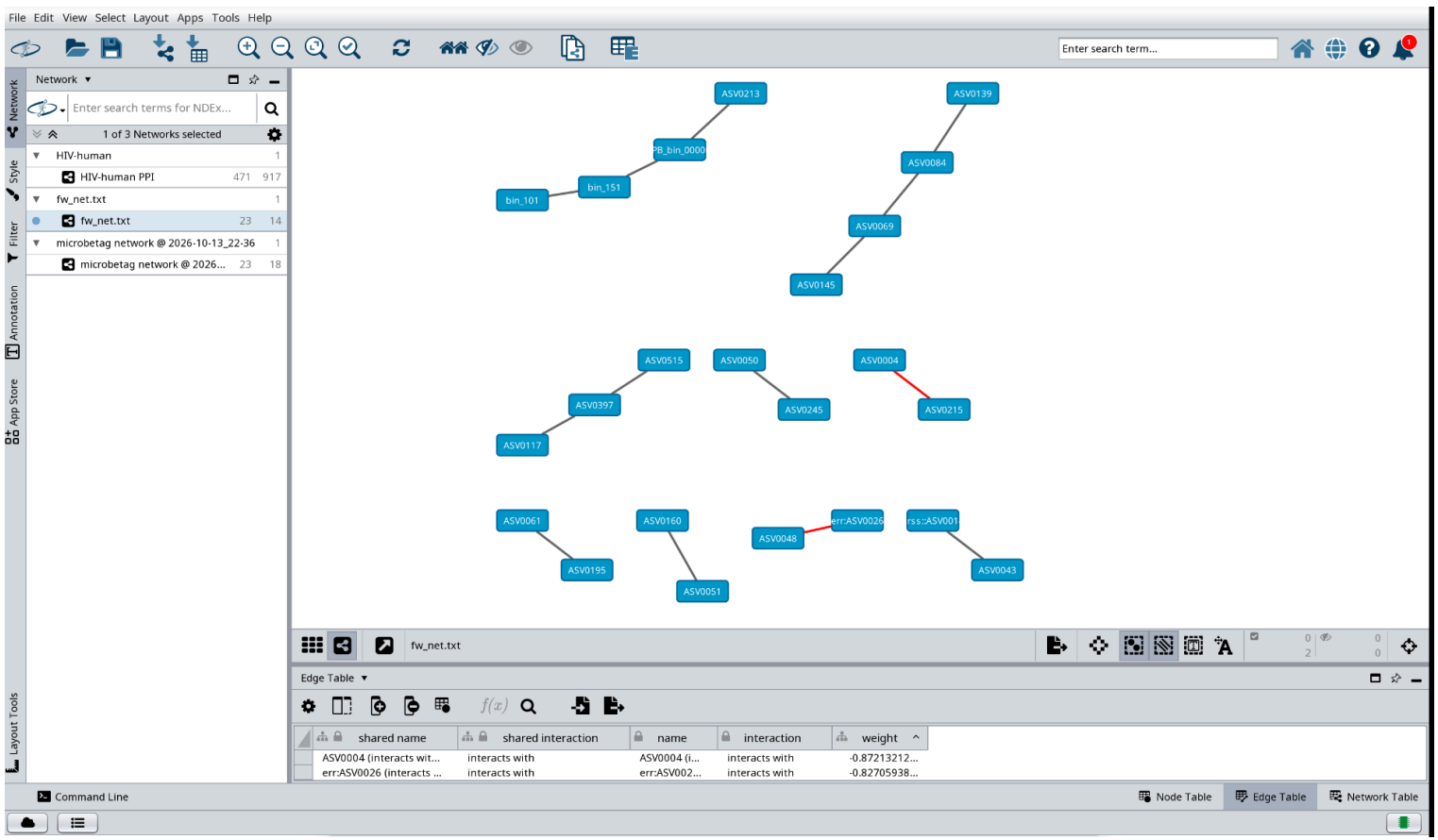
Making sense of a hairball is hard

Amplicon / metagenomic surveys → **co-occurrence networks**: thousands of **undirected** edges between taxa that appear together (or avoid each other).



Who is working with *who* and *how*?

A co-occurrence network in Cytoscape



microbetag: a hypothesis generator

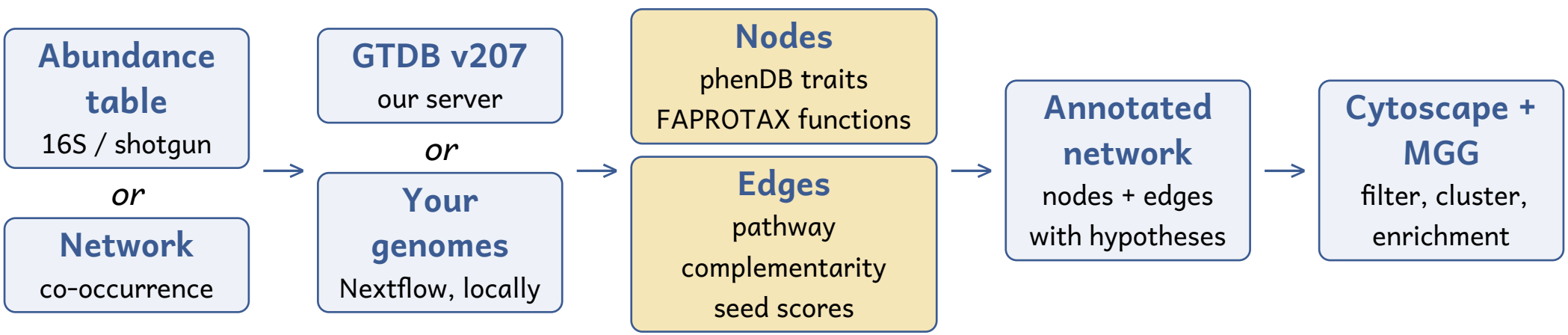
We cannot prove interactions from co-occurrence / mutual exclusion alone, but genomes can suggest **why** two taxa show such a pattern:

- *complementary* pathways → **cross-feedings**?
- overlapping needs → **competition**?
- shared traits → same **niche**?

▹ microbetag

Turns the hairball into **testable hypotheses**

How microbetag works



Node level: what are a species' phenotypic traits?

Each taxon gets **functional traits**: genome-based predictions (phenotrex) and literature-based **FAPROTAX** functions (105 in total), e.g.:

▹ Carbon

- methanogenesis
- fermentation
- cellulolysis

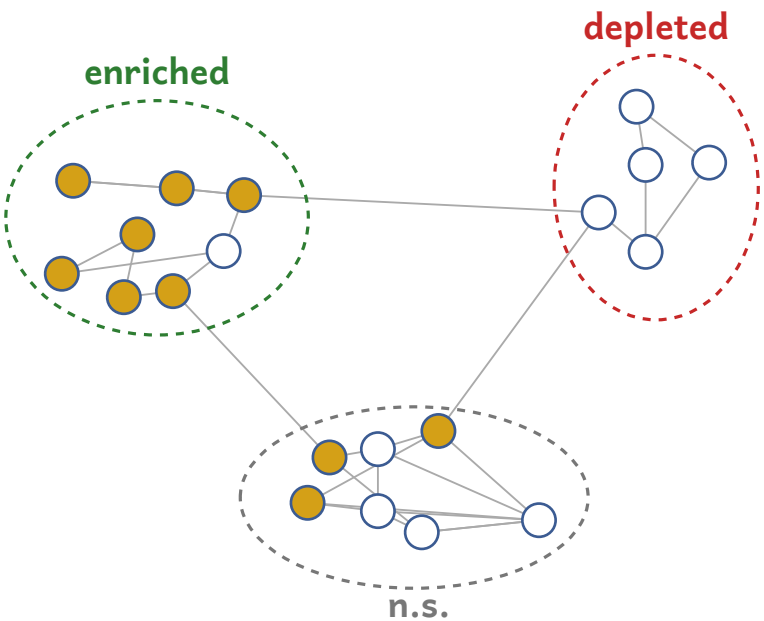
▹ Nitrogen & sulfur

- nitrogen fixation
- denitrification
- sulfate respiration

▹ Lifestyle

- obligate anaerobes
- human gut
- plant pathogen

Clusters & enrichment *analysis*



- taxon **with** the trait
- taxon **without**

1. **Cluster** the network (e.g. manta)
2. Per cluster & trait: **hypergeometric test** (wiki)
3. FDR correction → **enriched** / **depleted**

Each cluster reads as a **functional guild**

Edge level: what could they exchange?

▹ Pathway complementarity

KEGG modules one taxon misses and its partner can complete

▹ Seed scores

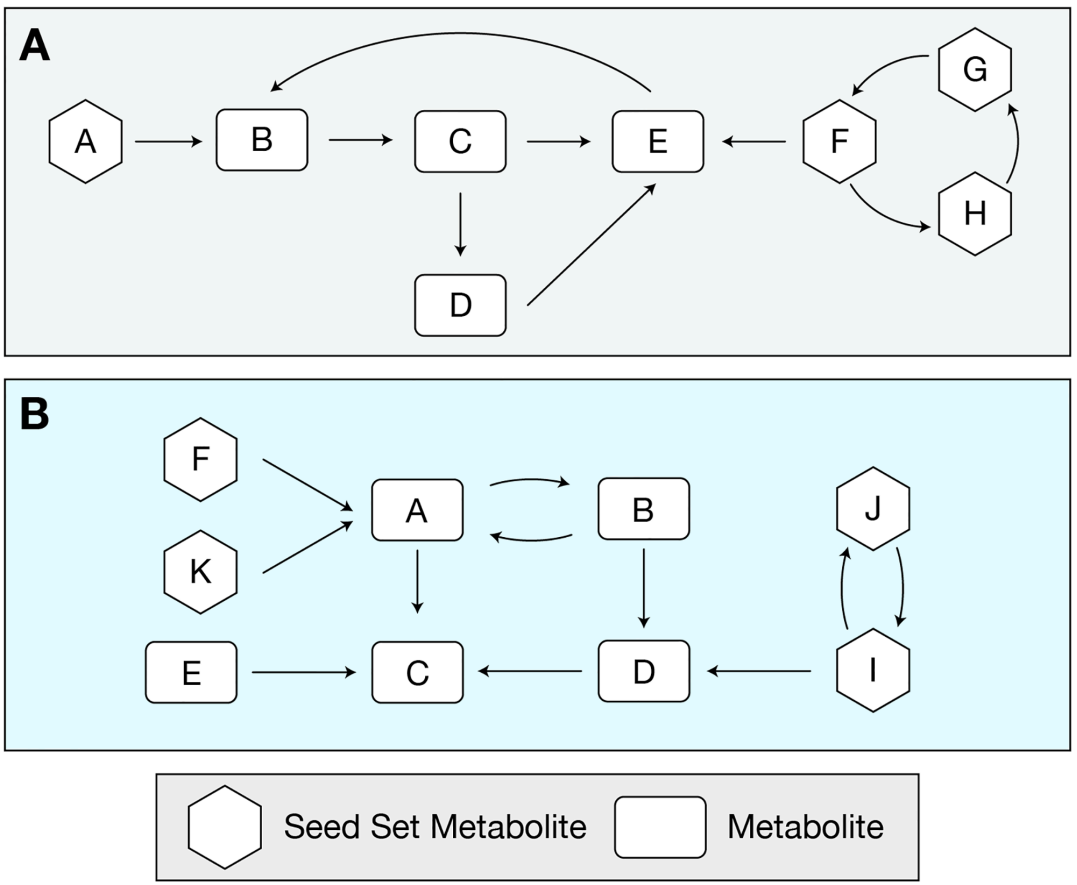
Competition and cooperation potential from metabolic network seed sets

▹ Seed complementarity

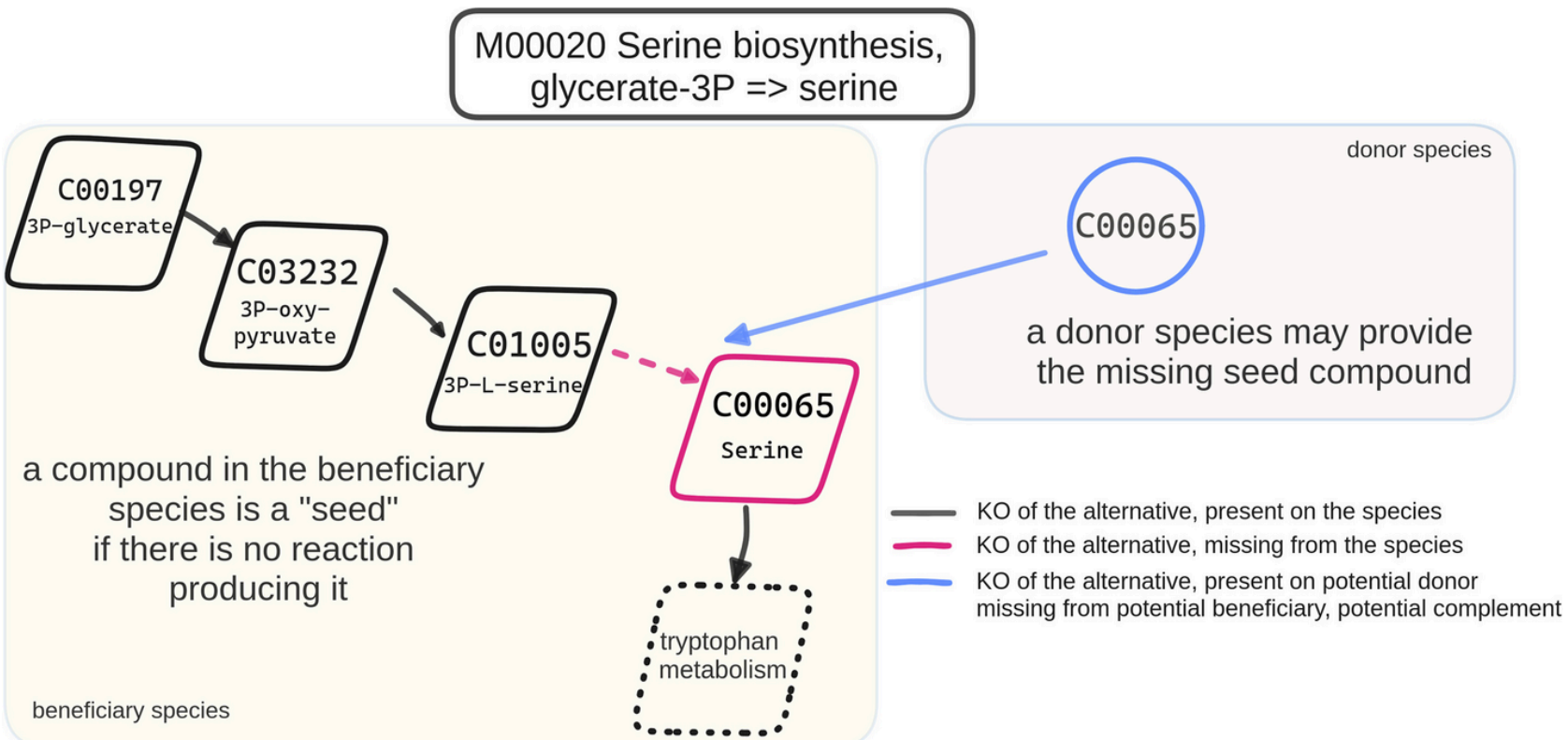
Candidate exchanged compounds from genome-scale models

The seed concept

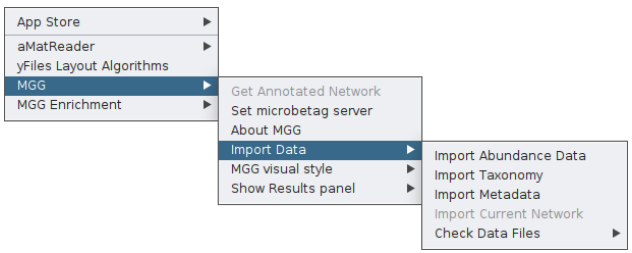
Pseudo metabolitc networks of species A and B



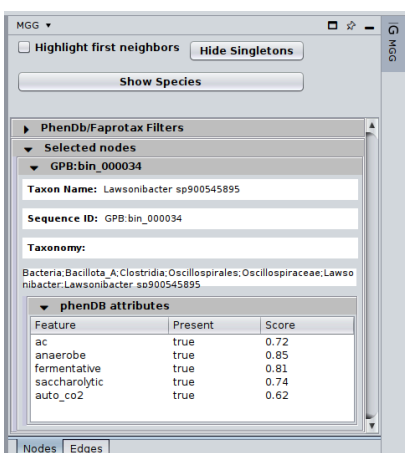
Seed complementarity



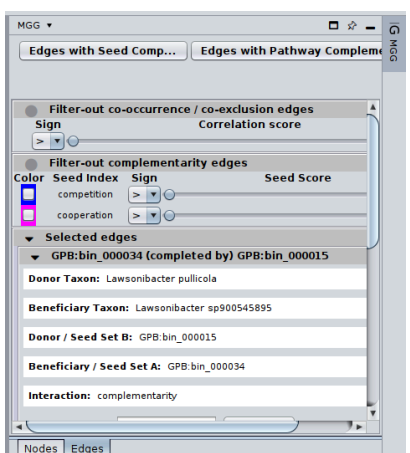
Parsing the network with MGG



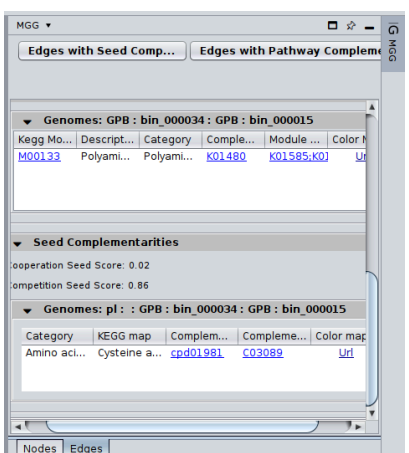
1. **Import** abundance table, taxonomy, metadata
2. **Get annotated network** from our server, or import your local one
3. Click *a* **node** or **edge** to read its annotations, **filter** edges by correlation or seed scores
4. Run **MGG Enrichment** on clusters



Nodes: taxonomy, phenDB / FAPROTAX

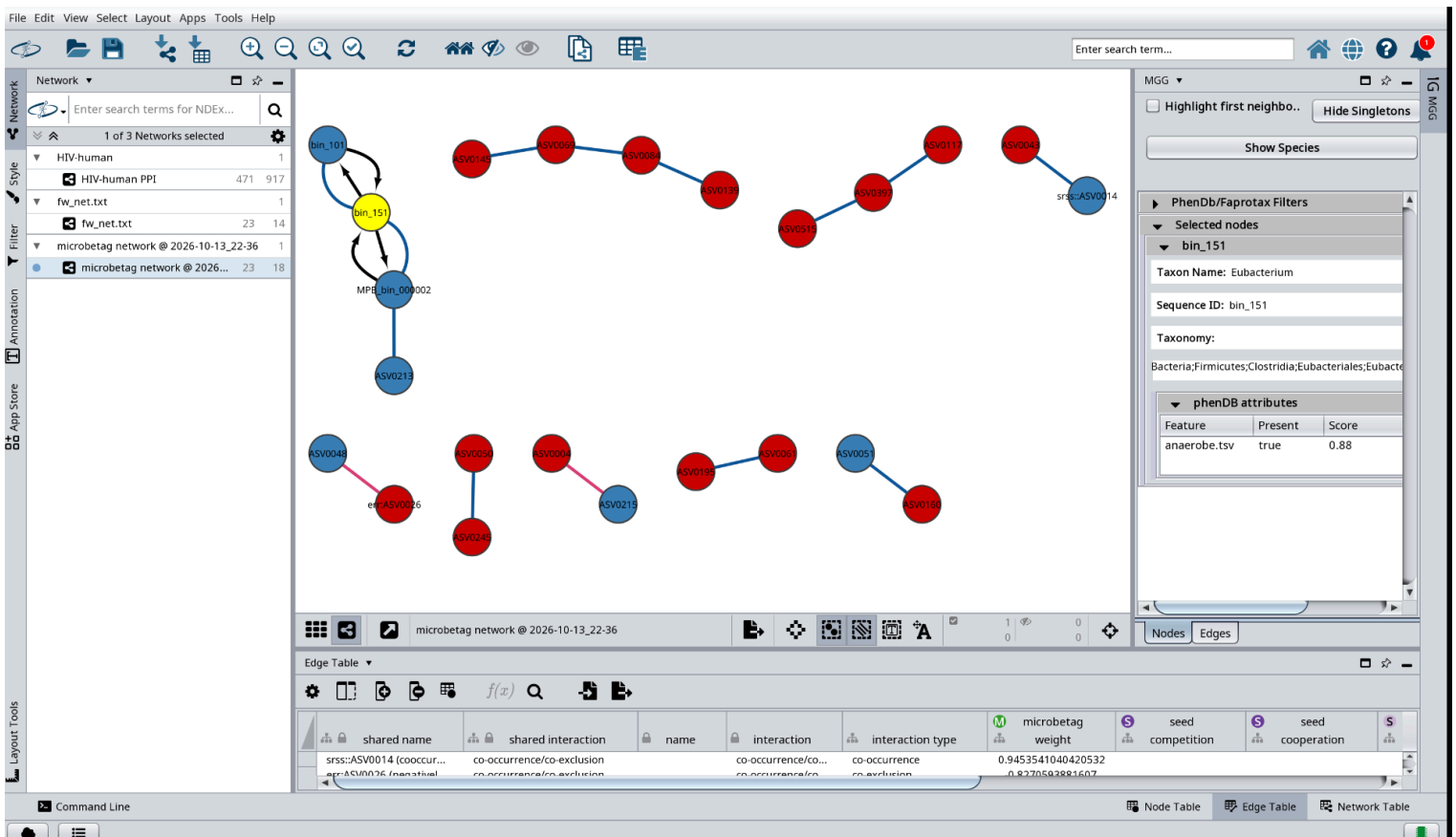


Edges: filters, donor → beneficiary



Edges: KEGG modules, seed scores

An annotated network in Cytoscape



Two ways to run microbetag

▹ On our server

- Against **GTDB v207** genomes
- Submit your **abundance table** or **network** through the MGG Cytoscape app
- Results come back as an annotated network

▹ Locally, with your genomes

- New: **Nextflow** pipeline
- Use your own genomes / MAGs
- Import the annotated network into Cytoscape with MGG

Either way: explore, filter and cluster the annotated network in **Cytoscape**.

microbetag.readthedocs.io · github.com/msysbio/microbetag

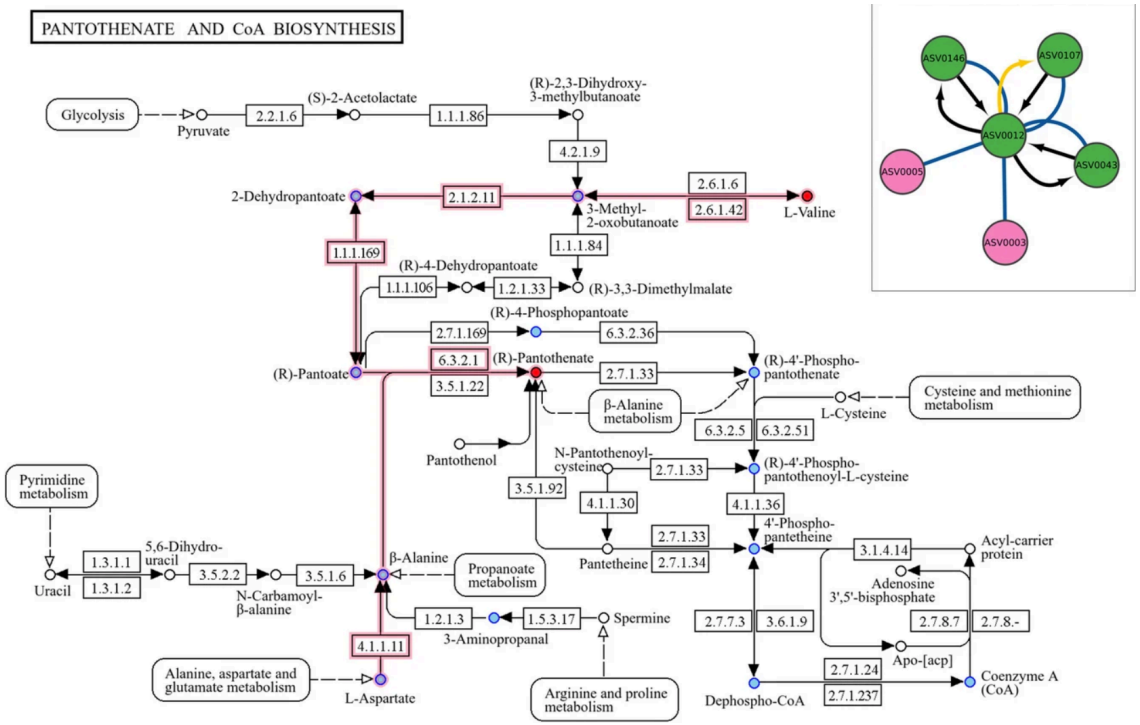
Case study: infant gut

Cabrera *et al.* (2019)
iron supplementation in infants

E. lenta → *F. plautii*
ASV0107 donor ASV0012 beneficiary

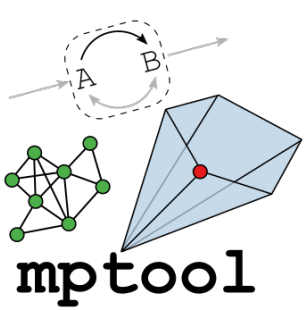
Complements: L-valine, (R)-pantothenate
→ vitamin B5 → **coenzyme A**

Hypothesis: FOS boosts butyrate & *Flavonifractor* partly via **cross-feeding partners**



Limits — and what's next

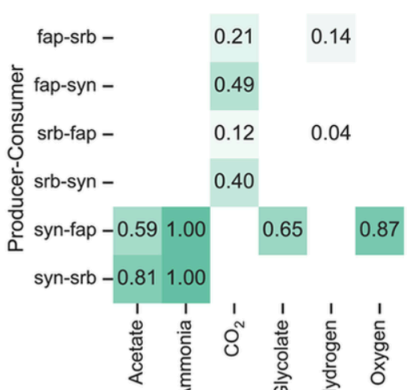
- Hypotheses, not proof: needs experimental validation
- Inherits the network's biases (depth, rare taxa, scale)
- Genome coverage: unmapped taxa stay unannotated
- Next: **interactions' dynamics** — do predicted interactions show up under different media ?



+



→



Take-home

▹ microbetag

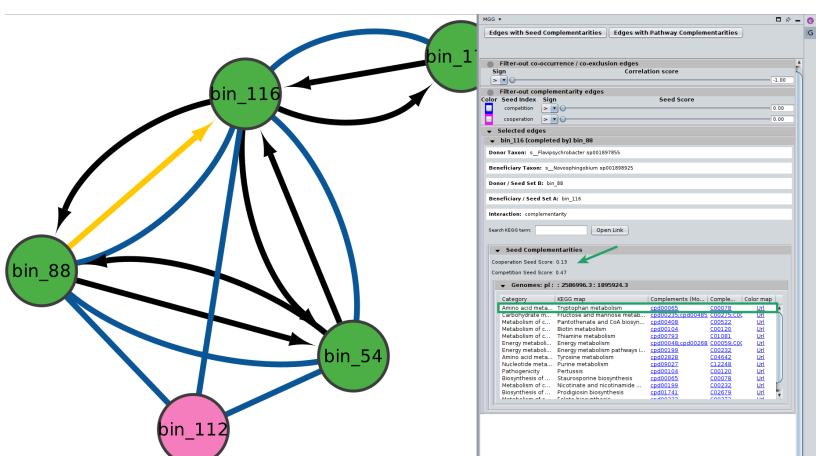
Makes sense of the hairball: from **who is with whom** to testable hypotheses of **what they do and exchange**.

Cytoscape app

Read the docs

Issues & feedback (Matrix)

Read the paper



Acknowledgements



Prof. Karoline Faust
Microbial Systems Biology lab
KU LEUVEN



John "Scooter" Morris
RBVI, UC San Francisco
NRNB



Ermis Delopoulos
MGG development
KU LEUVEN

ETH zürich **testing the software**



Aline Schneider



Annelies Geirnaert



Andrea Erega

Funding: 3D'omics (EU Horizon) · Google Summer of Code with NRNB

Questions?

microbetag on GitHub

Get these slides

I'd be happy to take your questions – or get in touch later 👋

charalampos-zafeiropoulos@uni-jena.de

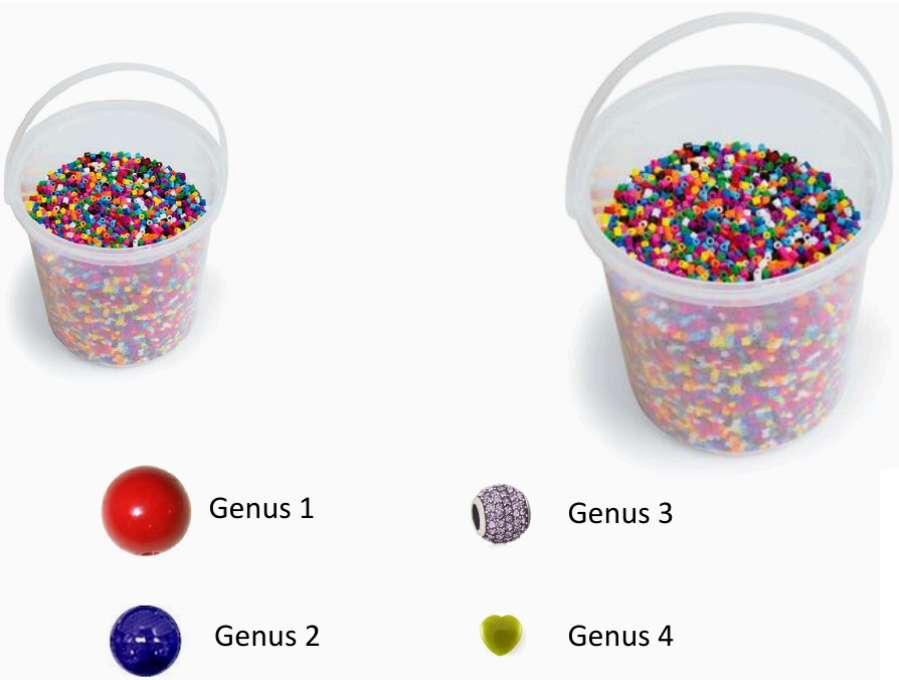
hariszaf.github.io

Challenges in co-occurrence inference

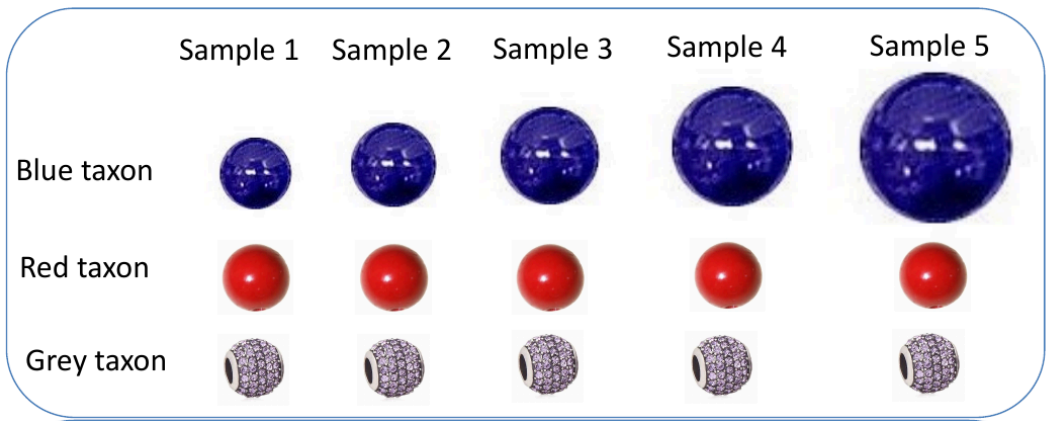
1. **Sequencing depth** varies across samples → compositional artefacts
2. **Rare taxa** → sparse counts, unstable correlations
3. **Indirect edges** → A–C may only reflect A–B and B–C; correlation ≠ interaction
4. **Undirected edges** → who affects whom? an edge has no direction
5. **Sampling scale** → interactions happen at the **microscale**, but we sample at the **macroscale**

Sampling scale

Technical variability.
Read count ≠ cell count



Compositionality



Absolute abundances

Relative abundances